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CHAPTER - 06

Foundations of Business Intelligence : Databases and Information Management.

1. What are the major problems associated with the traditional file environment?

* Data redundancy :

This occurs when there is duplicate data in multiple files.

* Data inconsistency :

This problem arises. When the same attribute has different values across various files.

* Program-data dependence :

This refers to the situation where changes in a program require corresponding changes to the data accessed by that program.

* Lack of flexibility :

Traditional file processing encourages specialized applications for each functional area, which limits the ability to quickly retrieve data for ad hoc queries or new reporting requirements.

* Poor Security :

Managing security across many separate files can be challanging - potentially leading to vulnarabilities.

* Lack of data sharing and availability.

Data is compartmentalized in separate files, it's difficult for different departments or users to share and access the same information efficiently.

02. Explain the difference between a primary key and a foreign key with a real-world example.

Primary key : A field within a table that is used to uniquely identify each record in that specific table. It ensures that every time entry in the table can be distinctively identified.

Foreign key : A foreign key is a primary key from one table that is used in a second table as a look-up field. Purpose is to identify records

from the original table, thereby establishing and maintaining relationships between different tables in a database.

Real - World example:

SUPPLIER table designed to store details about various suppliers. This table would have a field called supplier_Number. This serves as a primary key for the SUPPLIER table because each supplier has a unique number that identifies their record within that table.

Another table named PART, that stores information about different parts. To link each part to its supplier, the PART table would also include a field for Supplier - Number. In the PART table Supplier_Number acts as a foreign key.

03. How does a Database Management System (DBMS) improve data integrity and security compared to traditional file system?

1. Improved data integrity:

The data redundancy leads to data inconsistency. For instance, if different departments each maintain their own files, the same customer information might be stored in several places, and an update in one file might not be reflected in another. leading to unreliable information.

A DBMS fundamentally solves these problems. It controls redundancy and eliminates inconsistency by centralizing data.

Furthermore, database design processes involve normalization, which is the streamlining of complex data grouping to minimize redundant data elements and avoid an awkward many-to-many relationships. This ensures data are stored efficiently and logically.

2. Enhanced Security:

Poor Security → A DBMS provides significant advantage by ending enabling an organization to centrally manage data and data security. This allows for more robust security policies and easier implementation of access controls ensuring that only authorized users can access or modify specific data.

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04. What are the SELECT, JOIN, and PROJECT operations in a relational DBMS used for? Provide an example of each.

SELECT - Creates a subset of data from a table, including all records that meet standard criteria, essentially filters rows based on a condition.

Ex: If you have PART table, and you want to find all parts with Part_number 137 or 150, or you would use a SELECT operation.

```
SELECT * FROM PART WHERE Part_Number = 137 OR PART_number =
150;
```

JOIN: - Combines relational tables to provide a user more information than what is available in individual tables alone. Links two or more tables based on a common field typically a primary key and a foreign key relationship.

Ex. SUPPLIER Table - Primary Key : Supplier_Number

PART Table - Foreign key : Supplier_Number.

To findout which supplier provide which part you would use JOIN operation.

```
SELECT
    SUPPLIER.Supplier_number,
    SUPPLIER.SupplierName,
    PART.PartNumber,
    PART.PartName
FROM
    SUPPLIER
JOIN
    PART ON SUPPLIER.SupplierNumber = PART.Supplier_number;
```



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PROJECT - Creates a subset of columns from a table, resulting in new tables with only the specified information. (Column based filter)

Ex:

PART table - Part_number, Part_Name, Part_Color, Supplier_Number (attrib) and you only need a list of part-numbers and their names, you would use PROJECT operation:

SELECT Part_number, Part_Name FROM PART;

π Part-number, Part_Name (PART)

Q5. Compare and contrast relational databases and NoSQL databases. In what scenarios would you use each?

Relational Databases:

Organize data in the form of two-dimension tables. Each table contains data about an entity and its attributes.

Structure:

Arranges data in a grid of ~~the~~ column and rows. (tuples) Rows represent records for different entities, and fields (columns represent attributes.

Keys:

Primary Key, Foreign key.

Operations:

SELECT, JOIN, PROJECT

Data Integrity and Security

Scenarios for use:

1. Data is highly structured ~~and data integrity~~ and needs to conform to a predefined schema.
2. Strong consistency and data integrity are critical, with clear enforced relationships between different pieces of data.
3. Complex queries: involving multiple and precise joins are common.

Ex: Human resource database, financial system, and supply chain management

NoSQL databases:

represent a different approach to data management, often designed to overcome limitations of relational databases in specific contexts.

Data Model: They feature more flexible data model compared to the rigid tabular structure of relational databases.

Scalability: NoSQL databases are designed to store data sets across distributed machines, making them easier to scale. This horizontal scalability is crucial for handling "big data"

Data types: large volumes of unstructured and structured data.

Ex: Hadoop

Scenarios for use:

1. Data is unstructured or semi-structured and evolves rapidly.
2. No need to handle massive volumes of data that exceeds the capacity of traditional relational system.
3. High scalability and distributed processing
4. Flexibility

Ex: Web analytical, real-time big data applications, social media platform, IoT.

Contrast:

Feature	Relational Databases	NoSQL Databases
1. Data Model	Two dimensional tables, rigid, predefined schema	Flexibility data models; schema-less or dynamic schema.
2. Scalability	Can scale, but often more complex for massive, distributed data set. Scales vertically.	Designed for horizontal scalability.
3. Data type	structured data	unstructured and structured data

4. Consistency — Strong consistency often prioritizes availability and partition tolerance over immediate consistency.

06. What is data normalization? and why it is important in database design.

Streamlining complex groupings of data to minimize redundunt data elements and awkward many-to-many relationships.

Why it is important

1. Minimized redundancy and inconsistency:
2. Optimizes Data Storage and Efficiency:

By reducing redundancy, normalization ensures that the data is stored more efficiently.

After a normalization happens relation can be broken down into smaller, more efficient relations.

3. Enhanced Data integrity.

By having well normalized tables, it becomes easier to maintain the accuracy and consistency of data across the database.

4. Improves database performance and flexibility.

07. How does a cloud database differ from a traditional on-premise database system. Mention one advantage and one dicadvantage.

Functionality - Cloud database typically have less functionality than on-premise databases.

Ex: Amazon Relational Database Services and Microsoft SQL Azure

Traditional on-premise database systems, whether relational or NoSQL, are managed directly by the organization on their own hardware.

Advantage: Scalability

Dicadvantage - Dependang on Internet Connectivity.

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06. What is the role of data warehouse in business intelligence? How does it differ from a data mart.

Role of a Data warehouse in Business Intelligence :

- Stores current and historical data from many core operational transaction systems.
- It consolidate and standardized information for use across the entire enterprise. means data from various departments or systems is brought together and put into a consistent format, making it suitable for analysis.
- The data within a data warehouse cannot be altered.
- analysis and reporting tools.

How does Data Warehouse Differs from a Data Mart?

A data warehouse is designed to consolidate and standardize information for use across the entire enterprise.

A data mart, is a subset of data warehouse. It contains a summarized or focused portion of data that is specifically intended for use by a specific population of users. Typically a data mart focuses on a single subject or line of business.

09. Data mining and explain how businesses use it to make strategic decisions. Provide a real world example.

Data mining - the process of finding hidden patterns and relationships in large data sets, infers rules to predict future behavior.

Role in Strategic Decision Making.

1. understand customer behavior. - Can identify customer buying patterns. this helps businesses to tailor products, services, and marketing strategies more effectively.
2. Predict future trends :

Business can anticipate market changes, customer needs, and potential risks or opportunities.

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Improve business performance:

make better business decisions by concolidating, analyzing and providing access to data.

Ex: Retail Company.

Product placement, Targeted Promotions, Inventory Management.

10. What are some techniques used to ensure data quality in organization? Why is data quality critical for business performance?

Key techniques to Encure Data quality.

1. Control - redundancy & inconsistency.
2. Normalization
3. Referential Integrity.
4. Information Policy.
5. Data editing Cleansing.

Data quality Audit.

Why it matters.

Better decision.

Reduced Errors.

Business Succes.

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